# Chapter 14: Mendel and the Gene Idea

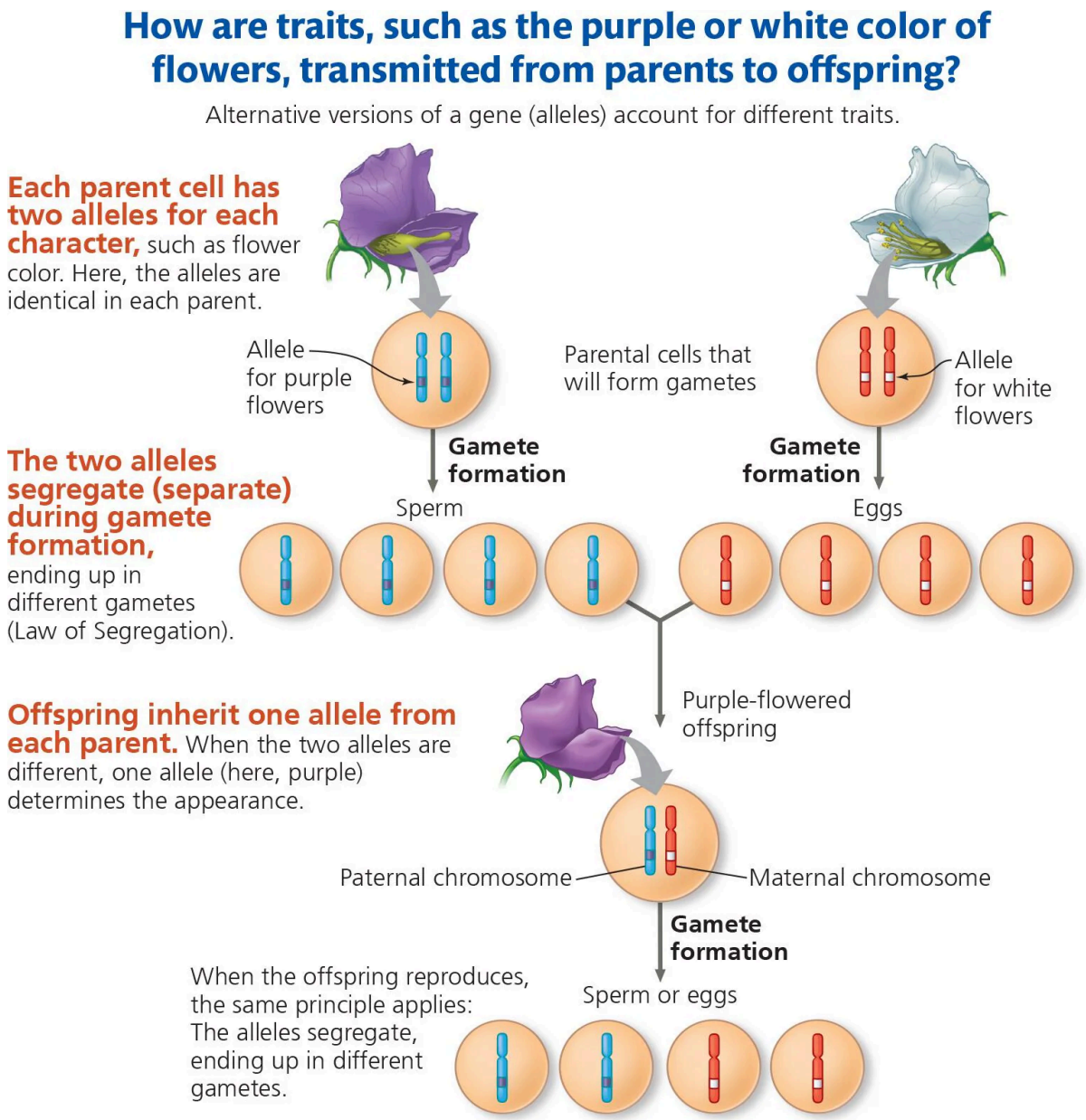
*Campbell Biology, AP Edition, 12e*

# Chapter 14

# Mendel and the Gene Idea

**Figure 14.1**

Flower color was one of the many characteristics of pea plants studied by Gregor Mendel. By breeding pea plants over many generations and carefully counting the types of offspring, Mendel developed a theory of inheritance that is the basis of modern genetics.




Each parent cell has two alleles for each character, such as flower color. Here, the alleles are identical in each parent flower. The two alleles segregate (separate) during gamete formation, ending up in different gametes (Law of Segregation). Offspring inherit one allele from each parent. When the two alleles are different, one allele (here, purple) determines the appearance. When offspring reproduce, the same principle applies: The alleles segregate, ending up in different gametes.

## Key Concepts

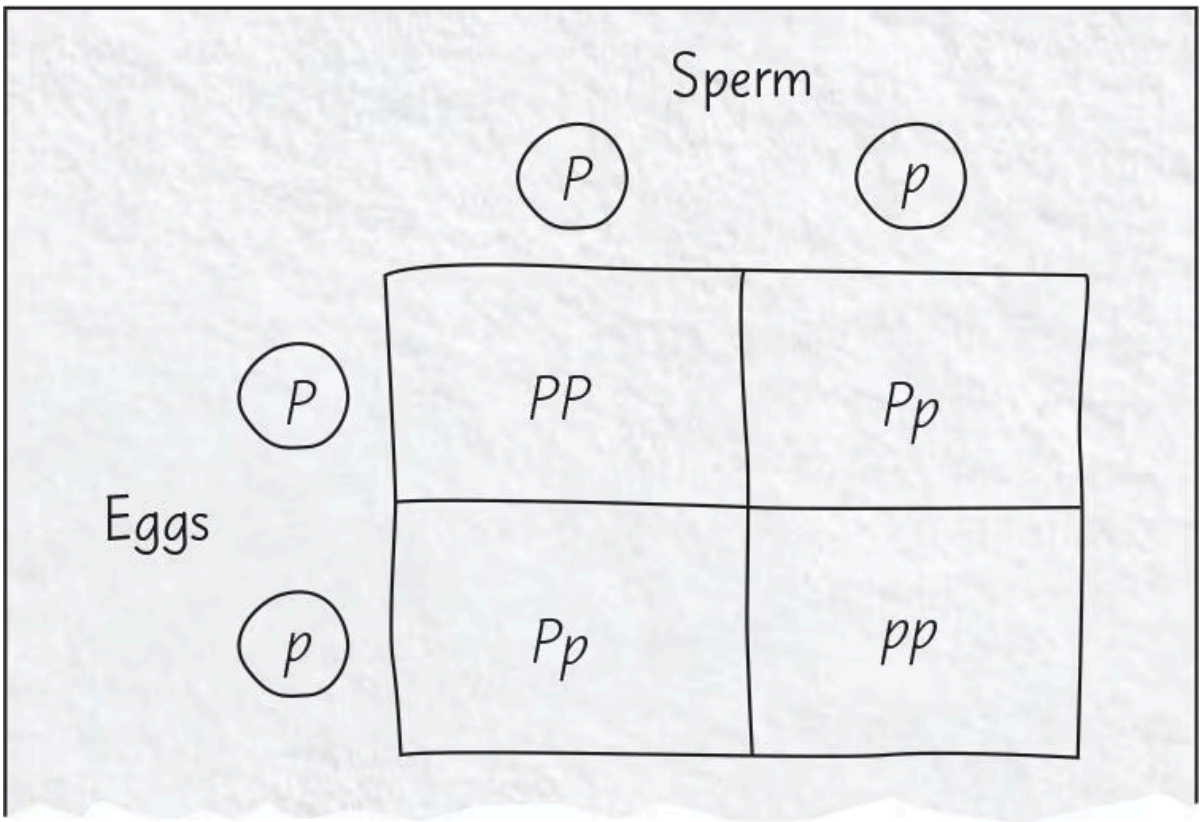
- 14.1**Mendel used the scientific approach to identify two laws of inheritance**
- 14.2**Probability laws govern Mendelian inheritance**

- 14.3**Inheritance patterns are often more complex than predicted by simple Mendelian genetics**
- 14.4**Many human traits follow Mendelian patterns of inheritance**

The meticulous work of Mendel and others in the 19th century provided evidence for a particulate mechanism of inheritance that led to our current understanding of the chromosomal basis of inheritance of genes from parent to offspring **(BIG IDEA 3)** and provided structural and functional evidence to support the relatedness of all domains **(BIG IDEA 1)**.

## Study Tip

**Solving genetics problems:** In Figure 14.5, you'll learn how to predict the offspring of a genetic cross with a Punnett square, like the example shown below. For crosses shown in other figures, cover up the Punnett square and draw your own. Also, check out the Tips for Genetics Problems at the end of the chapter.




A Punnett square illustrates the inheritance of alleles uppercase P and lowercase p, by depicting a square divided into 4 equal square sections. There are 2 rows and 2 columns forming 4 cells. To the left of the two rows is labeled Eggs, where the gamete with uppercase P (encircled) is placed near row 1 and the gamete with lowercase p (encircled) is placed near row 2. Above the columns is labeled sperm where the gamete with uppercase P (encircled) is placed near column 1 and the gamete with uppercase P (encircled) is placed near column 2. Top left quadrant is labeled by uppercase P uppercase P, top right quadrant is labeled uppercase p lowercase p, bottom left quadrant is labeled uppercase P lowercase p, and bottom right quadrant is labeled lowercase p lowercase p.

## Get Ready for This Chapter

In this chapter, you will learn about Mendelian genetics and more complex inheritance patterns, and will explore the rules of probability and the application of Mendelian principles to human genetics. Before beginning this chapter, you should be able to do the following:

- Describe the chromosomal makeup of a diploid cell (see Concept 13.2 and Figure 13.4).
- Describe how a gene codes for a protein (see Concept 5.5 and Figure 5.22).
- Explain how a protein's structure determines its function, using sickle-cell disease as an example (see Concept 5.4, Figure 5.18, and Figure 5.19).

The following questions provide a quick check of your basic knowledge in these areas.

1. What number and types of chromosomes are found in a somatic cell in an animal with a diploid number of 48?

24 chromosomes ( 12 from each parent, including one sex chromosome from each)

That's incorrect. Try again.

**Incorrect.** Review the following material: All somatic cells in animals have two sets of chromosomes, one haploid set (n) from each parent, and each set includes one sex chromosome. For an animal with a diploid number of 48 2n=48, the number of chromosomes in the somatic cells is 48 and the number of chromosomes in each haploid set is 24 n=24. Humans have a diploid number of 46 2n=46, and the number of chromosomes in each set is 23 2n=46, including one sex chromosome.

96 chromosomes ( 48 from each parent, including one pair of sex chromosomes from each)

That's incorrect. Try again.

**Incorrect.** Review the following material: All somatic cells in animals have two sets of chromosomes, one haploid set (n) from each parent, and each set includes one sex chromosome. For an animal with a diploid number of 48 2n=48, the number of chromosomes in the somatic cells is 48 and the number of chromosomes in each haploid set is 24 (n=24). Humans have a diploid number of 46 (2n=46), and the number of chromosomes in each set is 23 (n=23), including one sex chromosome.

23 chromosomes ( 11 from each parent, plus one sex chromosome)

That's incorrect. Try again.

Review the following material: All somatic cells in animals have two sets of chromosomes, one haploid set (n) from each parent, and each set includes one sex chromosome. For an animal with a diploid number of 48 (2n = 48), the number of chromosomes in the somatic cells is 48 and the number of chromosomes in each haploid set is 24 (n=24). Humans have a diploid number of 46 (2n=46), and the number of chromosomes in each set is 23 (n=23), including one sex chromosome.

48 chromosomes ( 24 from each parent, including one sex chromosome from each)

That's correct.

**Correct!**If you had trouble with this question, review the following material: All somatic cells in animals have two sets of chromosomes, one haploid set (n) from each parent, and each set includes one sex chromosome. For an animal with a diploid number of 48 (2n=48), the number of chromosomes in the

somatic cells is 48 and the number of chromosomes in each haploid set is 24 (n=24). Humans have a diploid number of 46 (2n=46), and the number of chromosomes in each set is 23 (n=23), including one sex chromosome.

2. The information contained in DNA is used to make which of the following product(s)?

```
tRNA only
```

That's incorrect. Try again.

**Incorrect.** Review the following material: The information that programs all of a cell's activities is encoded in the sequence of DNA nucleotides. DNA directs RNA synthesis and, through mRNA, directs the order of the amino acids during protein synthesis; this entire process is called gene expression. Two types of RNA required for protein synthesis are mRNA and tRNA. (Other RNAs, such as rRNAs, are also necessary, and these too are encoded by genes in DNA.)

```
proteins only
```

That's incorrect. Try again.

**Incorrect.** Review the following material: The information that programs all of a cell's activities is encoded in the sequence of DNA nucleotides. DNA directs RNA synthesis and, through mRNA, directs the order of the amino acids during protein synthesis; this entire process is called gene expression. Two types of RNA required for protein synthesis are mRNA and tRNA. (Other RNAs, such as rRNAs, are also necessary, and these too are encoded by genes in DNA.)

```
mRNA only
```

That's incorrect. Try again.

**Incorrect.** Review the following material: The information that programs all of a cell's activities is encoded in the sequence of DNA nucleotides. DNA directs RNA synthesis and, through mRNA, directs the order of the amino acids during protein synthesis; this entire process is called gene expression. Two types of RNA required for protein synthesis are mRNA and tRNA. (Other RNAs, such as rRNAs, are also necessary, and these too are encoded by genes in DNA.)

```
proteins, mRNA, and tRNA
```

That's correct.

**Correct!** If you had trouble with this question, review the following material: The information that programs all of a cell's activities is encoded in the sequence of DNA nucleotides. DNA directs RNA synthesis and, through mRNA, directs the order of the amino acids during protein synthesis; this entire process is called gene expression. Two types of RNA required for protein synthesis are mRNA and tRNA. (Other RNAs, such as rRNAs, are also necessary, and these too are encoded by genes in DNA.)

3. Normal hemoglobin is a tetramer, consisting of two molecules of beta-globin and two molecules of alpha-globin; normal hemoglobin molecules do not associate with each other. In sickle-cell disease, the change in a single amino acid results in mutant hemoglobin tetramers, which associate with each other and assemble into large fibers. Based on this information alone, what can we conclude about the changes in the structure of sickle cell hemoglobin as compared to normal hemoglobin?

```
altered primary and quaternary structure
```

That's correct.

**Correct!** If you had trouble with this question, review the following material: The primary structure of a protein is its sequence of amino acids, secondary structure is the result of hydrogen bonds between the repeating constituents of the polypeptide backbone, tertiary structure is the three-dimensional shape stabilized by interactions between side chains, and quaternary structure results when two or more polypeptides associate with each other. Normal hemoglobin tetramers do not associate with each other. The single amino acid change that causes sickle-cell disease alters the primary and quaternary structure of the protein, and as a result, the abnormal hemoglobin tetramers aggregate into chains. Not enough information is given to determine whether secondary and tertiary interactions are affected.

```
altered primary structure only
```

That's incorrect. Try again.

**Incorrect.** Review the following material: The primary structure of a protein is its sequence of amino acids, secondary structure is the result of hydrogen bonds between the repeating constituents of the polypeptide backbone, tertiary structure is the three-dimensional shape stabilized by interactions between side chains, and quaternary structure results when two or more polypeptides associate with each other. Normal hemoglobin tetramers do not associate with each other. The single amino acid change that causes sickle-cell disease alters the primary and quaternary structure of the protein, and as a result, the abnormal hemoglobin tetramers aggregate into chains. Not enough information is given to determine whether secondary and tertiary interactions are affected.

```
altered secondary and tertiary structure
```

That's incorrect. Try again.

**Incorrect.** Review the following material: The primary structure of a protein is its sequence of amino acids, secondary structure is the result of hydrogen bonds between the repeating constituents of the polypeptide backbone, tertiary structure is the three-dimensional shape stabilized by interactions between side chains, and quaternary structure results when two or more polypeptides associate with each other. Normal hemoglobin tetramers do not associate with each other. The single amino acid change that causes sickle-cell disease alters the primary and quaternary structure of the protein, and as a result, the abnormal hemoglobin tetramers aggregate into chains. Not enough information is given to determine whether secondary and tertiary interactions are affected.

```
altered quaternary structure only
```

That's incorrect. Try again.

**Incorrect.** Review the following material: The primary structure of a protein is its sequence of amino acids, secondary structure is the result of hydrogen bonds between the repeating constituents of the polypeptide backbone, tertiary structure is the three-dimensional shape stabilized by interactions between side chains, and quaternary structure results when two or more polypeptides associate with each other. Normal hemoglobin tetramers do not associate with each other. The single amino acid change that causes sickle-cell disease alters the primary and quaternary structure of the protein, and as a result, the abnormal hemoglobin tetramers aggregate into chains. Not enough information is given to determine whether secondary and tertiary interactions are affected.

## Learning Objectives

After studying this chapter, you should be able to do the following:

- 14.1Explain Mendel's two laws of inheritance.
- 14.2Describe the laws of probability and explain how they can be used to solve complex genetics problems.
- 14.3Use examples to show how genetic inheritance patterns can be affected by complete dominance, incomplete dominance, codominance, multiple alleles, pleiotropy, epistasis, and polygenic inheritance.

- 14.4Explain what a pedigree is and use examples to show how human traits follow Mendelian patterns of inheritance.

---